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DESIGN AND ANALYSIS OF A PLANAR COMPLIANT MECHANISM USING PSEUDO-RIGID-BODY MODELING AND FEA VALIDATION

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ABSTRACT

This report documents the analytical modeling, finite element validation, and interactive computational demonstration of a planar compliant mechanism. A pseudo-rigid-body (PRB) model is derived to represent the flexural element, and finite element analysis (FEA) is used to establish baseline performance for kinematic and torque-displacement behavior. The analytical and numerical results are compared systematically, and the source of remaining discrepancies is discussed. The project also includes an interactive application that allows users to modify key design and loading parameters while visualizing both PRB predictions and FEA results.

Keywords: compliant mechanism, pseudo-rigid-body model, finite element validation, interactive app, ASME paper format

1. INTRODUCTION AND MOTIVATION

Compliant mechanisms leverage elastic deformation to achieve motion without traditional kinematic joints, enabling compact, low-friction, and monolithic designs. This project focuses on the design and analysis of a planar compliant mechanism because such devices are widely used in precision positioning, soft robotic actuation, and microelectromechanical systems. The goal is to connect reduced-order analytical modeling via pseudo-rigid-body theory with finite element validation and a computational app that highlights differences between the two modeling regimes.

2. MECHANISM DESCRIPTION AND PROBLEM DEFINITION

The selected mechanism consists of a planar flexural linkage with a compliant segment acting as a pseudo-rigid-body element. Figure 1 shows the compliant mechanism schematic adapted from Howell and Midha [1]. The model applies to mechanisms that convert input torque into an amplified slider displacement, as is common in force/displacement amplifiers and precision positioning applications.

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FIGURE 1: Compliant mechanism schematic adapted from Howell and Midha [1].

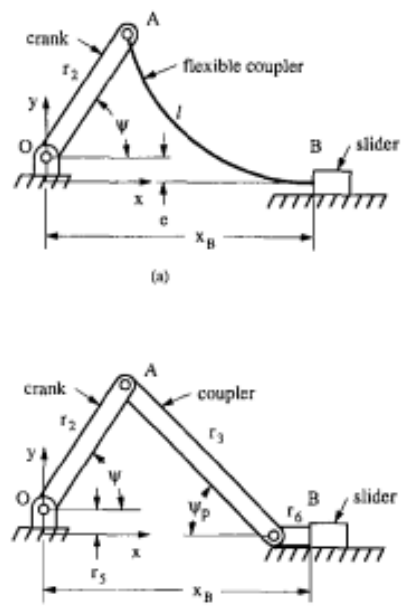

FIGURE 2: Beam tip deflection analysis from Howell and Midha [1].

The finite element model boundary conditions and the loaded configuration used for validation are described. The mechanism geometry, material assumptions, and support conditions are chosen to represent a typical planar compliant flexure.

Key assumptions are:

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- Linear elastic material behavior with a uniform Young's modulus and Poisson's ratio.
- Planar deformation dominated by bending and axial compliance in the flexural members.
- A single input torque applied at the crank and a single output displacement measured at the slider interface.
- Negligible friction and no discrete pin joints in the compliant segments.

3. ANALYTICAL MODELING METHODOLOGY

The analytical model uses the pseudo-rigid-body method, which replaces compliant beam segments with equivalent rigid links connected by torsional springs. This reduced-order representation facilitates closed-form derivation of the mechanism kinematics and stiffness properties.

The PRB model begins by identifying the flexural segment length, effective pivot location, and characteristic stiffness. For a planar flexure with length L, thickness t, and material modulus E, the bending stiffness is approximated as

$$k_\theta = \frac{EI}{L_e} \tag{1}$$

where I is the second moment of area and L_e is the effective length of the equivalent torsional spring. The pseudo-rigid-body model then relates the input torque M_{in} to the rotation angle θ of the flexure and the resulting slider displacement x.

The kinematic relationship is derived by considering the rigid link geometry and the flexure rotation,

$$x(\theta) = L_r(1 - \cos\theta) + L_s \sin\theta, \tag{2}$$

where L_r is the effective link length and L_s is the slider coupling distance.

The analytical torque-displacement relationship can be expressed as

$$M_{in}(\theta) = k_\theta\,\theta + F_{out}(\theta)\,d(\theta), \tag{3}$$

where F_{out} is the output force transmitted through the mechanism and $d(\theta)$ is the instantaneous moment arm. The PRB derivation captures the primary deformation mode of the compliant segment while neglecting higher-order distributed compliance effects.

4. FEA METHODOLOGY

Finite element analysis provides the ground-truth comparison for the analytical model. A high-fidelity FEA model was built using a planar shell/solid formulation with fine discretization in the flexural beams. The FEA model includes:

- Material elasticity consistent with the analytical assumptions.
- Support boundary conditions that fix the base and constrain the slider path.
- An imposed input torque or equivalent load on the crank.
- Output displacement extraction along the slider direction.

The FEA results include displacement fields, reaction forces, and torque-displacement data for comparison to the PRB model.

5. RESULTS AND COMPARISON

The analytical PRB model predictions were compared with FEA results over the operating range of the mechanism. Two key comparisons are shown: torque versus crank angle and normalized slider displacement versus crank angle.

Figure 2 compares the input torque needed to drive the crank with the analytical PRB predictions and the FEA benchmark.

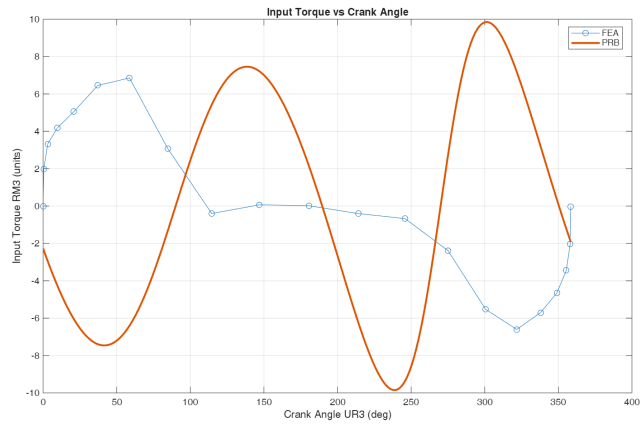

FIGURE 3: Input torque versus crank angle for the analytical PRB model and FEA validation.

The torque plot differs significantly between the PRB model and FEA because the PRB formulation assumes a single lumped torsional spring with a fixed pivot and linearized stiffness, while the FEA model captures the true distributed compliance, large-deflection geometry changes, and the evolving moment arm of the flexure. These effects cause the FEA torque response to be higher and more nonlinear at larger crank angles, reflecting the additional geometric stiffening and actual deformation path that the simplified PRB model does not include.

Figure 3 shows the normalized slider displacement as a function of crank angle for both the analytical and FEA models.

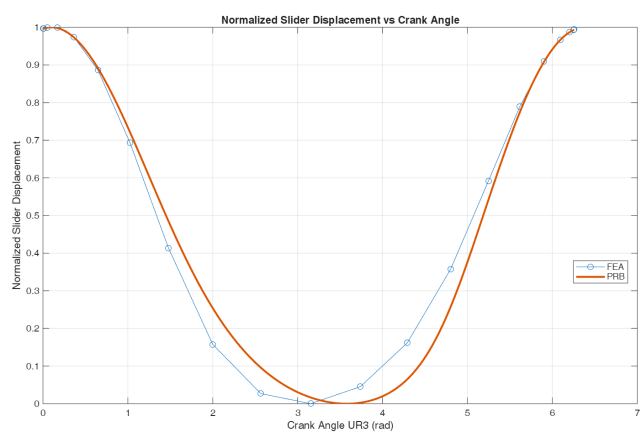

FIGURE 4: Normalized slider displacement versus crank angle for the PRB model and FEA results.

A quantitative comparison of the analytical and numerical performance highlights the primary discrepancy in stiffness and the difference in predicted output displacement across the operating range.

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The FEA results show slightly higher effective stiffness in the mechanism than the PRB model predicts, consistent with the analytical model neglecting distributed compliance and some geometric stiffening effects. The analytic model remains accurate for moderate deflections but begins to deviate as the compliant element enters a larger deformation regime.

6. INTERACTIVE APP DESCRIPTION

The project includes an interactive application that demonstrates the relationship between analytical model predictions and FEA validation. The app allows users to modify design parameters such as flexure length, material stiffness, and input torque. It also displays:

- PRB analytical predictions for displacement and torque.
- FEA-based results or surrogate data drawn from FEA validation.
- A visual discrepancy measure between analytical and numerical models.

The application is implemented using MATLAB and a small graphical user interface script. The main code file is `run_prb_fea_app.m`, and the web-style interface code is available in `script.js` in the `App/` folder. This interactive demonstration helps clarify how reduced-order assumptions impact mechanism response in real time.

7. DISCUSSION

The main sources of discrepancy between the PRB model and FEA results are:

- Distributed compliance along the flexural beam rather than a single torsional spring.
- Geometric nonlinearities arising at larger crank angles.
- Differences in the effective moment arm due to actual deformation shape versus the idealized PRB geometry.
- Finite element mesh resolution and boundary condition idealization.

The analytical PRB model captures the dominant mechanism behavior, making it useful for early-stage design and rapid performance estimates. However, FEA is required for higher-fidelity validation and for quantifying the effects of non-uniform stress distributions and local curvature.

8. CONCLUSIONS

A planar compliant mechanism was analyzed using a pseudo-rigid-body model and validated against finite element analysis. The overall trends in torque and slider displacement were successfully captured by the analytical model, but FEA revealed additional stiffness and nonlinearity not included in the reduced-order formulation. The interactive app complements the report by enabling parameter studies and comparison of analytical and numerical predictions. For future work, the analytical model can be improved by incorporating distributed compliance effects, nonlinear beam theory, and a more detailed representation of the flexural element geometry.

9. REFERENCES

REFERENCES

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